

Retrieval-Grounded and Uncertainty-Aware AI for Surgical Video Intelligence

Bloomberg Data Science Ph.D. Fellowship 2027-2028 | Mohamed Sameh Guessoum

Surgical videos contain rich procedural evidence, but most AI systems for surgical-video analysis still stop at isolated predictions such as phase labels or tool detections. Clinicians and researchers often need more than a label: they need evidence-linked timelines, reliable retrieval of relevant moments, structured summaries, and clear signals of uncertainty. My proposed research develops retrieval-grounded and uncertainty-aware AI methods for surgical video intelligence, using cataract surgery as the initial domain.

The project addresses a central problem in trustworthy multimodal AI: how can a system connect video evidence, temporal events, generated reports, and user questions without hallucinating or overstating confidence? Cataract surgery is a strong setting for this work because surgical videos have structured phases, visible instruments, clinically meaningful events, and natural documentation needs. Through my research internship at Massachusetts Eye and Ear / Harvard Medical School, I have experience working with cataract-surgery video datasets including Cataract-101, Cataract-1K, CATARACTS, and CaDIS, while distinguishing public datasets from restricted or institutionally authorized data. My current pipeline uses convolutional and temporal models for phase recognition, object detection models for tool detection, and retrieval-grounded methods for interactive report generation and question answering.

Aim 1 will build an evaluation benchmark for grounded surgical-video retrieval and reporting. The benchmark will organize video segments, phase labels, tool detections, event annotations, and report-style outputs into a unified evaluation structure. Instead of measuring only aggregate classification accuracy, the benchmark will measure temporal localization, retrieval precision, evidence coverage, report faithfulness, and uncertainty calibration. This will make it possible to ask whether a generated answer is supported by the correct video moments and whether the system knows when evidence is insufficient.

Aim 2 will develop a retrieval-grounded multimodal architecture for surgical-video question answering and report generation. The system will first identify candidate evidence segments using phase recognition, tool detection, event cues, and learned video representations. A retrieval layer will then select relevant temporal evidence before a generation component produces structured summaries or answers. The model will be constrained to cite or otherwise link outputs to retrieved evidence. This design is intended to reduce hallucination by separating evidence selection from language generation.

Aim 3 will add uncertainty-aware and human-centered decision support. The system will estimate uncertainty at multiple levels: phase and tool predictions, event localization, retrieval relevance, and generated report confidence. When uncertainty is high, the output should flag the need for human review rather than providing unsupported conclusions. I will evaluate performance across dataset source, procedure phase, visual quality, annotation granularity, and distribution shift. The final system will support clinician-facing workflows such as reviewing a surgical case, locating relevant events, and generating draft documentation that remains tied to inspectable evidence.

This work is relevant to Bloomberg's data science interests because it advances several general methods beyond medicine: retrieval-augmented multimodal reasoning, document-style generation from structured and unstructured evidence, trustworthy AI evaluation, and human-AI collaboration. The same technical questions arise whenever systems must retrieve evidence from complex multimodal records and produce reliable summaries for expert users. By grounding outputs in verifiable evidence and measuring uncertainty, this research aims to make generative AI systems more useful in high-stakes analytical workflows.

My preparation combines optimization, machine learning, and applied biomedical AI. My M.S. thesis at Worcester Polytechnic Institute develops an adaptive multi-agent system for data-driven heuristic orchestration in 1D/2D bin-packing problems, extended toward industrial 3D bin packing. That work trained me to design systems that coordinate specialized methods, validate constraints, and adapt to problem structure. My biomedical AI work includes cataract-surgery video analysis, multimodal cardiology triage with explainable AI, and medical-image segmentation. As an incoming Computer Science PhD student at WPI, I plan to develop AI systems that combine strong empirical performance with transparency, uncertainty awareness, and practical usefulness for expert decision-making.